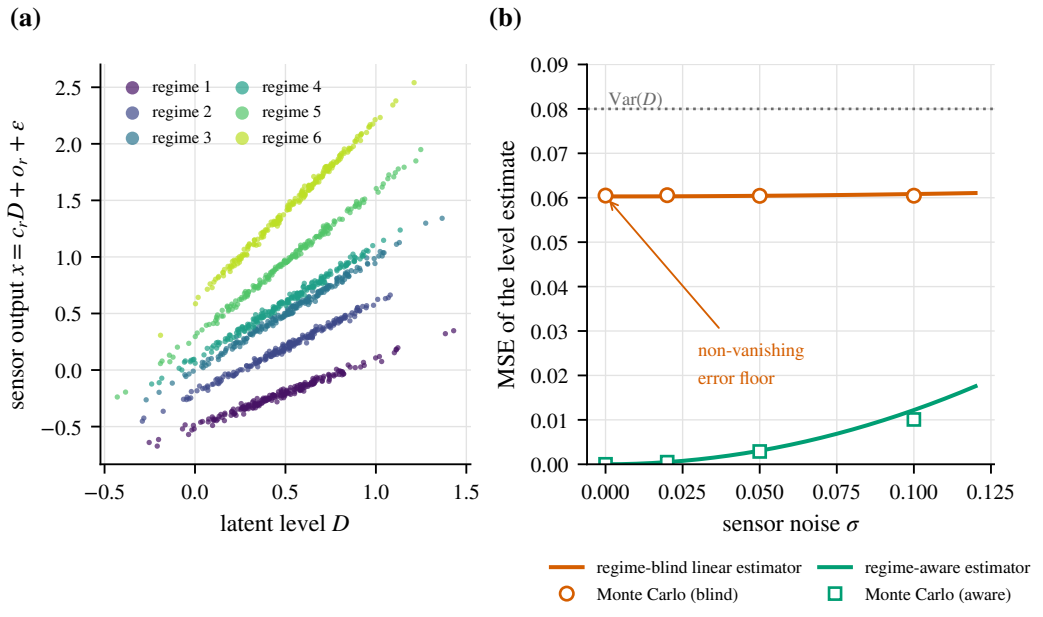
(a)
regime 1
regime 2
regime 3
regime 4
regime 5
regime 6
sensor output $x = c_r D + o_r + \varepsilon$
latent level D
(b)
Var(D)
non-vanishing
error floor
MSE of the level estimate
sensor noise σ
regime-blind linear estimator
regime-aware estimator
Monte Carlo (blind)
Monte Carlo (aware)